PRAKTIKUM BASIS DATA

03_SCRIPTING_SQL_BASIC_TP

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TUGAS PENDAHULUAN

Soal Tugas Pedahuluan :

1. Buatlah sebuah table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.

```
SQL> CREATE TABLE mahasiswa (
  2  id NUMBER PRIMARY KEY,
  3  nama VARCHAR2(50),
  4  tempat_lahir VARCHAR2(30),
  5  tanggal_lahir DATE,
  6  nomor_hp VARCHAR2(15),
  7  email VARCHAR2(50),
  8  tinggi_badan NUMBER,
  9  berat_badan NUMBER
 10  );
CREATE TABLE mahasiswa (
```

2. Lakukan insert data pada table nomor 1 sebanyak 15 rows.

```
SQL> INSERT INTO mahasiswa VALUES (3,'Andi Saputra','Surabaya',TO_DATE('10-11-2001','DD-MM-YYYY'),'083456789123','andi@gmail.com',172,68);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (4,'Rina Putri','Semarang',TO_DATE('05-01-2003','DD-MM-YYYY'),'084567891234','rina@gmail.com',158,50);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (1,'Budi Santoso','Bandung',TO_DATE('21-08-2001','DD-MM-YYYY'),'081234567891','budi@gmail.com',170,65);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (2,'Siti Aminah','Jakarta',TO_DATE('15-03-2002','DD-MM-YYYY'),'082345678912','siti@gmail.com',160,55);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (5,'Dedi Kurniawan','Yogyakarta',TO_DATE('12-07-2002','DD-MM-YYYY'),'085678912345','dedi@gmail.com',175,70);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (6,'Lina Marlina','Malang',TO_DATE('20-02-2001','DD-MM-YYYY'),'086789123456','lina@gmail.com',162,52);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (7,'Agus Setiawan','Solo',TO_DATE('18-06-2002','DD-MM-YYYY'),'087891234567','agus@gmail.com',168,64);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (8,'Dina Rahmawati','Bandung',TO_DATE('22-09-2003','DD-MM-YYYY'),'088912345678','dina@gmail.com',159,53);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (9,'Rudi Hartono','Medan',TO_DATE('11-04-2002','DD-MM-YYYY'),'089123456789','rudi@gmail.com',174,72);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (10,'Tina Kartika','Bogor',TO_DATE('07-12-2001','DD-MM-YYYY'),'081234567800','tina@gmail.com',161,54);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (11,'Farhan Hakim','Depok',TO_DATE('14-10-2002','DD-MM-YYYY'),'081345678900','farhan@gmail.com',173,67);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (12,'Salsa Nabila','Bekasi',TO_DATE('09-03-2003','DD-MM-YYYY'),'082456789011','salsa@gmail.com',157,49);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (13,'Iqbal Ramadhan','Padang',TO_DATE('30-05-2001','DD-MM-YYYY'),'083567890122','iqbal@gmail.com',176,71);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (14,'Maya Sari','Lampung',TO_DATE('02-02-2002','DD-MM-YYYY'),'084678901233','maya@gmail.com',160,56);
1 row created.
SQL> INSERT INTO mahasiswa VALUES (15,'Hendra Wijaya','Palembang',TO_DATE('17-08-2003','DD-MM-YYYY'),'085789012344','hendra@gmail.com',178,75);
1 row created.
```

3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:
   a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)
      - LOWER/UPPER

```
SQL> SELECT LOWER(nama) FROM mahasiswa;

LOWER(NAMA)
--------------------------------------------------
andi saputra
rina putri
budi santoso
siti aminah
dedi kurniawan
lina marlina
agus setiawan
dina rahmawati
rudi hartono
tina kartika
farhan hakim

LOWER(NAMA)
--------------------------------------------------
salsa nabila
iqbal ramadhan
maya sari
hendra wijaya

15 rows selected.
```

- INITCAP

```
SQL> SELECT INITCAP(nama) FROM mahasiswa;

INITCAP(NAMA)
--------------------------------------------------
Andi Saputra
Rina Putri
Budi Santoso
Siti Aminah
Dedi Kurniawan
Lina Marlina
Agus Setiawan
Dina Rahmawati
Rudi Hartono
Tina Kartika
Farhan Hakim

INITCAP(NAMA)
--------------------------------------------------
Salsa Nabila
Iqbal Ramadhan
Maya Sari
Hendra Wijaya

15 rows selected.
```

- CONCAT

```
SQL> SELECT CONCAT(nama, tempat_lahir) FROM mahasiswa;

CONCAT(NAMA,TEMPAT_LAHIR)
-------------------------------------------------------------------------
----------
Andi SaputraSurabaya
Rina PutriSemarang
Budi SantosoBandung
Siti AminahJakarta
Dedi KurniawanYogyakarta
Lina MarlinaMalang
Agus SetiawanSolo
Dina RahmawatiBandung
Rudi HartonoMedan
Tina KartikaBogor
Farhan HakimDepok

CONCAT(NAMA,TEMPAT_LAHIR)
-------------------------------------------------------------------------
----------
Salsa NabilaBekasi
Iqbal RamadhanPadang
Maya SariLampung
Hendra WijayaPalembang

15 rows selected.
```

- LENGTH

```
SQL> SELECT nama, LENGTH(nama) FROM mahasiswa;

NAMA                                               LENGTH(NAMA)
-------------------------------------------------- ------------
Andi Saputra                                                 12
Rina Putri                                                   10
Budi Santoso                                                 12
Siti Aminah                                                  11
Dedi Kurniawan                                               14
Lina Marlina                                                 12
Agus Setiawan                                                13
Dina Rahmawati                                               14
Rudi Hartono                                                 12
Tina Kartika                                                 12
Farhan Hakim                                                 12

NAMA                                               LENGTH(NAMA)
-------------------------------------------------- ------------
Salsa Nabila                                                 12
Iqbal Ramadhan                                               14
Maya Sari                                                     9
Hendra Wijaya                                                13

15 rows selected.
```

- SUBSTR

```
SQL> SELECT SUBSTR(nama,1,4) FROM mahasiswa;

SUBSTR(NAMA,1,4)
----------------
Andi
Rina
Budi
Siti
Dedi
Lina
Agus
Dina
Rudi
Tina
Farh

SUBSTR(NAMA,1,4)
----------------
Sals
Iqba
Maya
Hend

15 rows selected.
```

- LPAD/RPAD

```
SQL> SELECT LPAD(nama,10,'*') FROM mahasiswa;

LPAD(NAMA,10,'*')
----------------------------------------
Andi Saput
Rina Putri
Budi Santo
Siti Amina
Dedi Kurni
Lina Marli
Agus Setia
Dina Rahma
Rudi Harto
Tina Karti
Farhan Hak

LPAD(NAMA,10,'*')
----------------------------------------
Salsa Nabi
Iqbal Rama
*Maya Sari
Hendra Wij

15 rows selected.
```

- LTRIM

```
SQL> SELECT LTRIM('   Budi') FROM dual;

LTRI
----
Budi
```

- RTRIM

```
SQL> SELECT RTRIM('Budi   ') FROM dual;

RTRI
----
Budi
```

- INSTR

```
SQL> SELECT nama, INSTR(nama,'a') FROM mahasiswa;

NAMA                                               INSTR(NAMA,'A')
-------------------------------------------------- ---------------
Andi Saputra                                                     7
Rina Putri                                                       4
Budi Santoso                                                     7
Siti Aminah                                                     10
Dedi Kurniawan                                                  11
Lina Marlina                                                     4
Agus Setiawan                                                   10
Dina Rahmawati                                                   4
Rudi Hartono                                                     7
Tina Kartika                                                     4
Farhan Hakim                                                     2

NAMA                                               INSTR(NAMA,'A')
-------------------------------------------------- ---------------
Salsa Nabila                                                     2
Iqbal Ramadhan                                                   4
Maya Sari                                                        2
Hendra Wijaya                                                    6

15 rows selected.
```

- REPLACE

```
SQL> SELECT REPLACE(nama,'a','@') FROM mahasiswa;

REPLACE(NAMA,'A','@')
-----------------------------------------------
Andi S@putr@
Rin@ Putri
Budi S@ntoso
Siti Amin@h
Dedi Kurni@w@n
Lin@ M@rlin@
Agus Seti@w@n
Din@ R@hm@w@ti
Rudi H@rtono
Tin@ K@rtik@
F@rh@n H@kim

REPLACE(NAMA,'A','@')
-----------------------------------------------
S@ls@ N@bil@
Iqb@l R@m@dh@n
M@y@ S@ri
Hendr@ Wij@y@

15 rows selected.
```

b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH|YEAR]))

- MM/MON/MONTH

```
SQL> SELECT tanggal_lahir,
  2  TO_CHAR(tanggal_lahir,'MM') AS bulan
  3  FROM mahasiswa;

TANGGAL_L BU
--------- --
10-NOV-01 11
05-JAN-03 01
21-AUG-01 08
15-MAR-02 03
12-JUL-02 07
20-FEB-01 02
18-JUN-02 06
22-SEP-03 09
11-APR-02 04
07-DEC-01 12
14-OCT-02 10

TANGGAL_L BU
--------- --
09-MAR-03 03
30-MAY-01 05
02-FEB-02 02
17-AUG-03 08

15 rows selected.
```

- DD/DY

```
SQL> SELECT tanggal_lahir,
  2  TO_CHAR(tanggal_lahir,'DD') AS hari
  3  FROM mahasiswa;

TANGGAL_L HA
--------- --
10-NOV-01 10
05-JAN-03 05
21-AUG-01 21
15-MAR-02 15
12-JUL-02 12
20-FEB-01 20
18-JUN-02 18
22-SEP-03 22
11-APR-02 11
07-DEC-01 07
14-OCT-02 14

TANGGAL_L HA
--------- --
09-MAR-03 09
30-MAY-01 30
02-FEB-02 02
17-AUG-03 17

15 rows selected.
```

- YYYY/YY/RR

```
SQL> SELECT tanggal_lahir,
  2  TO_CHAR(tanggal_lahir,'YYYY') AS tahun
  3  FROM mahasiswa;

TANGGAL_L TAHU
--------- ----
10-NOV-01 2001
05-JAN-03 2003
21-AUG-01 2001
15-MAR-02 2002
12-JUL-02 2002
20-FEB-01 2001
18-JUN-02 2002
22-SEP-03 2003
11-APR-02 2002
07-DEC-01 2001
14-OCT-02 2002

TANGGAL_L TAHU
--------- ----
09-MAR-03 2003
30-MAY-01 2001
02-FEB-02 2002
17-AUG-03 2003

15 rows selected.
```

- AM/PM/HH/HH24

```
SQL> SELECT TO_CHAR(SYSDATE,'AM') FROM dual;

TO
--
AM
```

- MI/SS

```
SQL> SELECT TO_CHAR(SYSDATE,'SS') FROM dual;

TO
--
20
```

- ADD_MONTHS(d,n)

```
SQL> SELECT ADD_MONTHS(tanggal_lahir,6) FROM mahasiswa;

ADD_MONTH
---------
10-MAY-02
05-JUL-03
21-FEB-02
15-SEP-02
12-JAN-03
20-AUG-01
18-DEC-02
22-MAR-04
11-OCT-02
07-JUN-02
14-APR-03

ADD_MONTH
---------
09-SEP-03
30-NOV-01
02-AUG-02
17-FEB-04

15 rows selected.
```

- SYSDATE

```
SQL> SELECT SYSDATE FROM dual;

SYSDATE
---------
10-MAR-26
```

- LAST_DAY(d)

```
SQL> SELECT LAST_DAY(tanggal_lahir) FROM mahasiswa;

LAST_DAY(
---------
30-NOV-01
31-JAN-03
31-AUG-01
31-MAR-02
31-JUL-02
28-FEB-01
30-JUN-02
30-SEP-03
30-APR-02
31-DEC-01
31-OCT-02

LAST_DAY(
---------
31-MAR-03
31-MAY-01
28-FEB-02
31-AUG-03

15 rows selected.
```

- NEXT_DAY(d,day_name)

```
SQL> SELECT NEXT_DAY(tanggal_lahir,'MONDAY') FROM mahasiswa;

NEXT_DAY(
---------
12-NOV-01
06-JAN-03
27-AUG-01
18-MAR-02
15-JUL-02
26-FEB-01
24-JUN-02
29-SEP-03
15-APR-02
10-DEC-01
21-OCT-02

NEXT_DAY(
---------
10-MAR-03
04-JUN-01
04-FEB-02
18-AUG-03

15 rows selected.
```

- MONTHS_BETWEEN(d1,d2)

```
SQL> SELECT MONTHS_BETWEEN(SYSDATE,tanggal_lahir) FROM mahasiswa;

MONTHS_BETWEEN(SYSDATE,TANGGAL_LAHIR)
-------------------------------------
                                  292
                            278.17275
                           294.656621
                           287.850169
                           283.946943
                           300.688879
                           284.753395
                           269.624363
                           286.979201
                           291.108234
                           280.882427

MONTHS_BETWEEN(SYSDATE,TANGGAL_LAHIR)
-------------------------------------
                           276.043718
                           297.366298
                           289.269524
                           270.785653

15 rows selected.
```

- ROUND(d,[MONTH| YEAR])

```
SQL> SELECT ROUND(tanggal_lahir,'MONTH') FROM mahasiswa;

ROUND(TAN
---------
01-NOV-01
01-JAN-03
01-SEP-01
01-MAR-02
01-JUL-02
01-MAR-01
01-JUL-02
01-OCT-03
01-APR-02
01-DEC-01
01-OCT-02

ROUND(TAN
---------
01-MAR-03
01-JUN-01
01-FEB-02
01-SEP-03

15 rows selected.

SQL> SELECT ROUND(tanggal_lahir,'YEAR') FROM mahasiswa;

ROUND(TAN
---------
01-JAN-02
01-JAN-03
01-JAN-02
01-JAN-02
01-JAN-03
01-JAN-01
01-JAN-02
01-JAN-04
01-JAN-02
01-JAN-02
01-JAN-03

ROUND(TAN
---------
01-JAN-03
01-JAN-01
01-JAN-02
01-JAN-04

15 rows selected.
```

c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))

- (TO_NUMBER (char)

```
SQL> SELECT TO_NUMBER('12345') FROM dual;

TO_NUMBER('12345')
------------------
             12345
```

- TO_CHAR (date)

```
SQL> SELECT TO_CHAR(SYSDATE,'DD-MM-YYYY') FROM dual;

TO_CHAR(SY
----------
10-03-2026
```

- TO_CHAR (number)

```
SQL> SELECT TO_CHAR(tinggi_badan) FROM mahasiswa;

TO_CHAR(TINGGI_BADAN)
--------------------------------------
172
158
170
160
175
162
168
159
174
161
173

TO_CHAR(TINGGI_BADAN)
--------------------------------------
157
176
160
178

15 rows selected.
```

- TO_DATE (char)

```
SQL> SELECT TO_DATE('21-08-2001','DD-MM-YYYY') FROM dual;

TO_DATE('
---------
21-AUG-01
```

d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

- (ABS(n)

```
SQL> SELECT ABS(-10) FROM dual;

  ABS(-10)
----------
        10
```

- MOD (m,n)

```
SQL> SELECT MOD(10,3) FROM dual;

 MOD(10,3)
----------
         1
```

- FLOOR (n)

```
SQL> SELECT FLOOR(12.7) FROM dual;

FLOOR(12.7)
-----------
         12
```

- POWER (m,n)

```
SQL> SELECT POWER(2,3) FROM dual;

POWER(2,3)
----------
         8
```

- ROUND (n[,m])

```
SQL> SELECT ROUND(12.456,2) FROM dual;

ROUND(12.456,2)
---------------
          12.46
```

- CEIL (n))

```
SQL> SELECT CEIL(12.2) FROM dual;

CEIL(12.2)
----------
        13
```